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Trust and Innovation Within the Firm: Evidence from Matched CEO-Firm Data

Contents

1 Big picture

- **Question.** Does a CEO's trust affect innovation inside the firm, and through what organizational mechanism does that effect operate?
- **Main claim.** A more trusting CEO increases firm innovation by building or reinforcing a trust-based culture that makes researchers more willing to pursue risky, exploratory R&D projects.
- **Data contribution.** The paper constructs a matched CEO–firm–patent data set covering 5,753 CEOs in 3,598 U.S. public firms from 2000 to 2011, linked to roughly 700,000 patent applications and more than 1 million inventors.
- **Trust measure.** A CEO's inherited generalized trust is inferred from the CEO's last name. The last name is mapped probabilistically to ethnic origins using deanonymized historical U.S. censuses, and those origins are linked to ethnic-origin trust measures from the General Social Survey.
- **Identification.** The main design uses CEO transitions and firm fixed effects. It asks whether the same firm produces more future patents under a more trusting CEO, after controlling for CEO and firm observables and year effects.
- **Baseline result.** A one-standard-deviation increase in CEO trust is associated with about 6% more future patent applications.
- **Patent-quality result.** The effect is driven by higher-quality innovation. CEO trust raises explorative patents, disruptive patents, patent importance, science-based patents, technological scope, average citations, and average patent value.
- **Mechanism.** Qualitative interviews and a conceptual model suggest that trusting managers tolerate failure. This encourages researchers to take high-risk, exploratory projects rather than safe exploitation projects.
- **Culture evidence.** Using Glassdoor-style employee review text classified by a fine-tuned BERT model, the paper shows that more trusting CEOs improve corporate trust culture, especially top-down trust from managers toward workers.
- **Selection channel.** More trusting CEOs select more trusting directors, especially new and staying directors, and less trusting directors are more likely to leave.
- **Boundary condition.** Trust is most productive when the firm's existing researcher pool is high quality. Excessive trust can be counterproductive when tolerance of failure is misplaced.
- **Within-CEO bilateral-trust test.** The paper also uses CEO fixed effects and bilateral trust toward inventors' countries. A CEO's higher bilateral trust toward a country raises patenting by inventors in that country, holding constant the CEO, country, and firm-year.
- **Takeaway.** Trust is not only a country-level cultural determinant of growth. It has a micro-foundation inside firms: a CEO's trust affects innovation by shaping risk-taking, culture, and the allocation of attention and authority toward researchers.

2 Data and measurement

2.1 Patent data and innovation measures

- The paper uses PATSTAT, which covers patent documents from more than 60 patent offices and contains information on application dates, publication dates, citations, technology classifications, patent families, and inventor addresses.

- To avoid double-counting, patents in the same patent family are treated as one invention and assigned to the year of the earliest application date.
- The baseline patent-count outcome is the number of patent applications filed by firm f in year $t+1$.
- Patent applications after 2011 are not used in the baseline because publication and citation lags make later patent data less complete.
- The paper also constructs multiple patent-quality measures: forward citations, patent value, explorative patents, disruptive patents, patent importance, scientific citations to non-patent literature, and technological scope.

Interpretation: Patent count measures the quantity of innovation, while quality-weighted outcomes test whether trust mainly increases low-value patents or instead encourages higher-risk and higher-quality innovation. The key finding is that the latter dominates.

2.2 CEO, firm, and baseline sample

- CEO and executive data come from BoardEx, which covers U.S. public firms and includes names, positions, dates, gender, education, and employment histories.
- Firm performance data come from Compustat.
- The baseline excludes financial firms and firms headquartered outside the United States.
- The final baseline sample has 3,598 U.S. public firms, 5,753 CEOs, and 29,384 firm-year-CEO observations from 2000 to 2011.
- About 60% of sample firms perform R&D and patent, and the sample firms file roughly 700,000 patent applications between 2001 and 2012.
- Each firm has on average 1.7 name-matched CEOs over the sample period; roughly two-thirds of firms have more than one CEO.

Interpretation: The baseline sample provides within-firm CEO turnover variation. This is essential because the empirical design asks how the same firm's innovation changes when a CEO with different inherited trust arrives.

2.3 CEO inherited-trust measure

The CEO-level generalized trust measure is

$$Trust_d = \sum_e w_{de}\, EthTrust_e, \tag{1}$$

where $EthTrust_e$ is average trust among U.S. residents with ethnic origin e, and w_{de} is the probability that CEO d descends from origin e.

- $EthTrust_e$ is computed using the General Social Survey trust question: "Generally speaking, would you say that most people can be trusted or that you can't be too careful in dealing with people?"
- To make the GSS comparison closer to CEOs, the paper focuses on GSS respondents in highly prestigious occupations.
- CEO last names are mapped to ethnic origins using full-count deanonymized U.S. censuses from 1910, 1920, 1930, and 1940.
- The census-based mapping uses 80 million individuals with foreign birthplace or ancestry and about 5 million unique last names.

- The paper restricts to last names with sufficient observations and allows a single name to map probabilistically to multiple ethnic origins.
- 83% of CEOs can be mapped to ethnic origins; mapped CEOs do not differ statistically from unmapped CEOs across observable characteristics.
- The three most common CEO ethnic origins are Irish, German, and English.

Interpretation: The measure is a proxy for inherited cultural norms rather than a direct survey of each CEO. Measurement error is likely, but the paper argues it should mainly attenuate estimated effects.

2.4 Validation and identifying variation in trust

- The CEO trust measure varies substantially across CEOs, as shown in Figure I.
- CEO trust is not predicted by firms' pre-transition patenting trends, as shown in Figure II.
- The paper also constructs alternative trust measures using LASSO, other GSS trust questions, and other surveys.
- Results are robust to controlling for other inherited cultural traits, including work ethic, risk preference, patience, and the CEO ethnic group's high-income share.

Interpretation: The empirical design requires CEO trust changes to be plausibly unrelated to pre-existing firm innovation trends. Figure II and robustness checks support this identifying assumption.

2.5 Employee reviews and corporate trust culture

- The paper uses online employee reviews from 2008 to 2017 to measure corporate trust culture.
- The author manually codes 10,000 reviews into categories of positive and negative trust sentiment.
- Trust direction is separated into top-down trust (managers trust workers), bottom-up trust (workers trust managers), and general trust.
- A fine-tuned BERT model classifies the full review sample into seven categories: positive top-down, positive bottom-up, positive general, negative top-down, negative bottom-up, negative general, and no trust information.
- The final culture sample contains 331,091 reviews across 266 firms and 397 CEO terms.

Interpretation: This text-analysis part tests whether CEO trust changes the firm's culture, not just patent counts. The results show that the effect is especially strong for top-down trust, the dimension most closely connected to tolerance of failure.

2.6 Bilateral trust and inventor-country data

- PATSTAT records inventor addresses, allowing the paper to identify the countries in which inventors reside.
- The bilateral-trust sample contains firm-by-inventor-country pairs for firms with patents by inventors in the country at least once between 2000 and 2012.
- The sample has 3,481 firm-country dyads, 730 firms with R&D labs in 27 countries, and 960 CEOs.
- A CEO's bilateral trust toward country c is constructed using the CEO's inherited origin and bilateral trust toward people from country c.

Interpretation: Bilateral trust enables a stronger within-CEO design: the same CEO may trust researchers from different countries differently, allowing CEO fixed effects to absorb all time-invariant CEO traits.

3 Conceptual framework and empirical specifications

3.1 Conceptual mechanism: trust, failure tolerance, and exploration

- The paper's conceptual framework views the manager as the principal and the researcher as the agent.
- Researchers choose between safer exploitation projects and riskier exploration projects.
- Exploration can fail because of bad luck or because the researcher is a bad type.
- A trusting manager places more prior probability on the researcher being a good type.
- After failure, a trusting manager is more likely to interpret the failure as bad luck and less likely to fire the researcher.
- Anticipating this tolerance of failure, good researchers choose more exploration.
- Exploration increases the probability of high-quality innovations, but it can also generate more failure.

Interpretation: Trust affects innovation not because it makes the CEO more optimistic about projects directly, but because it changes the researcher's incentive constraint. It lowers career risk from failure and encourages exploration.

3.2 Baseline within-firm specification

The main regression is

$$\text{arsinh}(pat_{f,t+1}) = \beta_1 Trust_d + \xi X_{ft} + \zeta Z_{dt} + \theta_f + \omega_t + \varepsilon_{fdt}. \tag{2}$$

- $pat_{f,t+1}$ is firm f's future patent-application count in year $t+1$.
- $Trust_d$ is CEO d's inherited generalized trust, standardized by the ethnicity-level standard deviation.
- X_{ft} includes firm controls, especially firm age and age squared.
- Z_{dt} includes CEO characteristics such as age, age squared, gender, education dummies, and tenure in the firm.
- θ_f are firm fixed effects.
- ω_t are year fixed effects.
- Standard errors are clustered by the CEO's main ethnicity, the level at which trust variation is measured.

Interpretation: The coefficient β_1 is identified from within-firm CEO transitions. It asks whether the same firm patents more after switching to a more trusting CEO.

3.3 Event-study version

The event-study specification compares firms before and after CEO transitions with different changes in CEO trust. Conceptually,

$$Y_{f,t} = \sum_{k \neq -1} \delta_k \, 1\{t - T_f = k\} \Delta Trust_f + \text{controls} + \text{fixed effects} + u_{ft}. \tag{3}$$

- T_f is the CEO transition year.

- $\Delta Trust_f$ is the change in trust from predecessor to successor CEO.
- Pre-transition coefficients test for differential pre-trends.
- Post-transition coefficients show the dynamic response of patenting.

Interpretation: Figures II and III are visual diagnostics for the common-trend assumption and the post-transition trust effect.

3.4 Quality-distribution regressions

To test the exploration mechanism, the paper replaces patent count with quality-weighted patent outcomes:

$$\text{arsinh}(QPat_{f,t+1}) = \beta_Q Trust_d + \xi X_{ft} + \zeta Z_{dt} + \theta_f + \omega_t + \varepsilon_{fdt}. \tag{4}$$

The outcomes include explorative patents, disruptive patents, patent importance, non-patent literature citations, technological scope, average citations, and average patent value.

Interpretation: The theory predicts effects concentrated in high-quality or explorative innovation. Table III and Figure V show precisely this pattern.

3.5 Trust-culture regressions

The corporate-culture specification is

$$ReviewTrust_{r,f,d,t} = \beta_C Trust_d + \Gamma Controls_{r,f,d,t} + \text{fixed effects} + \eta_{r,f,d,t}. \tag{5}$$

- The dependent variable takes values such as 100, -100, or 0 depending on whether a review expresses positive trust, negative trust, or no trust information.
- Separate outcomes consider both trust directions, top-down trust, bottom-up trust, and R&D-worker reviews.
- Observations are weighted so that each CEO term receives equal weight.

Interpretation: This specification tests whether the CEO's trust changes firm culture. The strongest results are for top-down trust and R&D-worker reviews.

3.6 Director selection specification

The director-selection regressions relate director trust to CEO trust:

$$DirectorTrust_{r,f,t,d} = \beta_D Trust_d + \text{controls} + \theta_f + \omega_t + \nu_{rfdt}. \tag{6}$$

- Columns separate new directors, staying directors, and leaving directors.
- Other columns use average director trust at the firm-year-CEO level.
- Directors who ever serve as CEO are excluded.

Interpretation: The goal is to test a top-down transmission channel: trusting CEOs may choose trusting directors, and less trusting directors may leave.

3.7 Right amount of trust: heterogeneity by researcher quality

The heterogeneity specification interacts CEO trust with pre-transition researcher-pool quality:

$$Y_{f,t} = \beta Trust_d + \lambda\big(Trust_d \times ResearcherQuality_f\big) + \text{controls} + \text{fixed effects} + e_{fdt}. \tag{7}$$

- Researcher quality is proxied by residual patent output before the CEO transition, after controlling for firm and CEO observables, industry fixed effects, and year fixed effects.
- Outcomes include all patents, top-quality patents, R&D efficiency, future patenting, sales, employment, and TFP.

Interpretation: Trust is productive when researchers are trustworthy. If the researcher pool is low quality, tolerance of failure may be misplaced and can reduce performance.

3.8 Bilateral-trust specification with CEO fixed effects

The within-CEO bilateral-trust regression is

$$\text{arsinh}(pat_{fc,t+1}) = \beta_2 BiTrust_{dc} + \omega_d + \kappa_c + \theta_{ft} + \varepsilon_{fdct}. \tag{8}$$

- $pat_{fc,t+1}$ is the patent count by inventors in country c working for firm f in year $t+1$.
- $BiTrust_{dc}$ measures CEO d's bilateral trust toward individuals from country c.
- ω_d are CEO fixed effects.
- κ_c are inventor-country fixed effects.
- θ_{ft} are firm-year fixed effects.
- Additional specifications add CEO-year, inventor-country-year, and firm-by-inventor-country fixed effects.

Interpretation: This is the paper's strongest design against unobserved CEO characteristics. It compares different R&D labs under the same CEO, asking whether labs in countries that the CEO trusts more patent more.

4 Identification and empirical design

4.1 Identifying assumption in the baseline design

- With firm fixed effects, identification comes from CEO changes within the same firm.
- The key assumption is that changes in CEO inherited trust are not driven by firm-specific innovation trends that would have occurred anyway.
- Figure II supports this by showing that pre-change patenting does not predict changes in CEO trust.
- Figure III supports the timing: patenting begins to move after CEO transitions rather than before them.

Interpretation: The baseline design is analogous to a difference-in-differences around CEO turnovers, but with a continuous treatment: the successor CEO's inherited trust.

4.2 Why the ethnic-origin trust measure helps

- Directly surveying CEOs about trust would be infeasible and potentially endogenous to current firm conditions.
- Inherited ethnic-origin trust is predetermined relative to the firm and CEO appointment.

- The measure captures cultural transmission and therefore provides a source of variation that is less directly affected by current firm innovation.

Interpretation: The strategy uses cultural inheritance as a proxy for CEO trust. It is imperfect, but its predetermined nature makes it useful for causal interpretation.

4.3 Addressing alternative explanations

The paper addresses many possible confounds:

- CEO demographics and education;
- tenure in the firm;
- other inherited cultural traits such as work ethic, patience, risk preference, and high-income share;
- CEO retirement and death transitions;
- country-of-origin controls;
- industry-specific trends;
- alternative patent transformations and count models;
- alternative estimators for staggered treatment effects;
- alternative trust measures.

Interpretation: Robustness checks are important because trust may correlate with many other cultural and personal attributes. The central result survives these checks.

4.4 Mechanism identification

- If CEO trust only increased R&D budgets, the paper should find higher R&D expenditure. It does not emphasize this as the main channel.
- If CEO trust encouraged exploration, the effect should concentrate in high-quality, explorative, and disruptive patents. It does.
- If CEO trust worked through culture, employee review text should show stronger trust culture. It does.
- If CEO trust cascaded through leadership selection, new and staying directors should be more trusting. They are.

Interpretation: The paper combines output, quality, culture, and personnel-selection evidence to triangulate the mechanism.

4.5 Remaining limitations

- The trust measure is a proxy and may contain substantial measurement error.
- CEO appointments are not random.
- Patent data miss non-patented innovation and may understate some forms of innovation.
- Employee-review data begin later and cover fewer firms than the patent sample.
- The bilateral-trust design is powerful but only applies to multinational R&D labs and inventor-country variation.

Interpretation: The paper's strength is the convergence of several research designs and datasets rather than a single perfectly randomized experiment.

5 Tables and figures

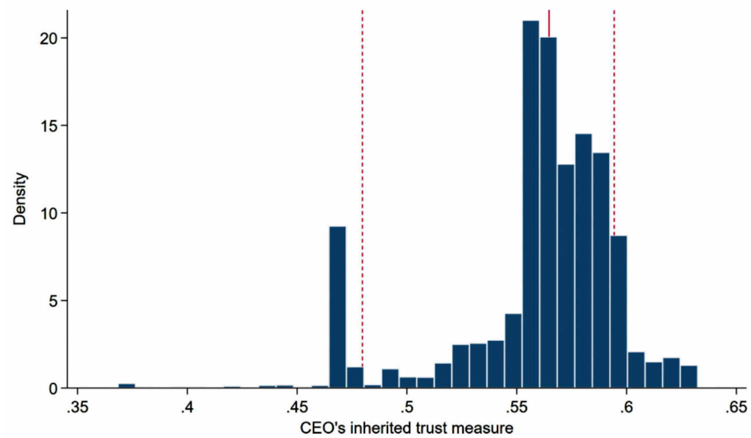
Visuals are directly extracted from the paper and grouped compactly. Tables IV, VI, and VII span two pages and are therefore shown in two panels.

5.1 Figure I: distribution of CEO inherited trust

Read:

- Figure I shows the 1st–99th percentile distribution of CEO inherited generalized trust.
- The distribution has substantial variation, creating meaningful cross-CEO differences.
- The average CEO trust measure is higher than the average trust measure in the general GSS sample but comparable to GSS respondents in prestigious occupations.

Economic message: CEO trust is not a binary trait. It varies continuously across CEOs, which allows the paper to estimate marginal effects of trust.

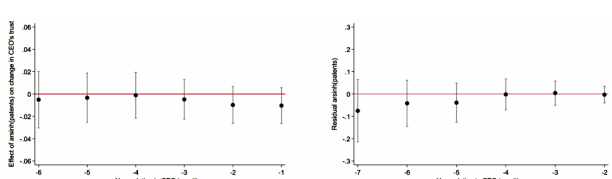


5.2 Figures II and III: pre-trends and dynamic effects

Read:

- Figure II, Panel A shows that pre-transition patenting does not predict the change in CEO trust around CEO turnovers.
- Figure II, Panel B shows average patenting around CEO transitions and supports a flat pre-trend.
- Figure III shows the main event-study pattern: patenting rises after a transition to a more trusting CEO.
- The visual evidence supports the idea that the effect appears after the CEO change rather than being driven by pre-existing innovation trends.

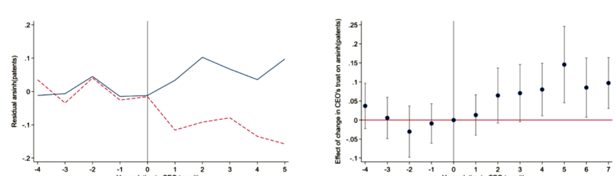
Economic message: These figures support the common-trend assumption behind the within-firm CEO-turnover design.



A. Common trends

B. Pre-trend

Figure II: pre-change patents and CEO trust



A. Patents by change in CEO's trust

B. Effects of change in CEO's trust

Figure III: pre- and post-change patent effects

5.3 Table I: baseline effect of CEO trust on patenting

Read:

- The baseline coefficient on CEO trust is 0.062, statistically significant at the 1% level.
- This implies that a one-standard-deviation increase in CEO trust is associated with about 6% more future patent applications.
- The effect remains similar after adding industry trends and country-of-origin controls.
- The effect is larger for forward-citation-weighted patents and patent-value-weighted patents.
- Retirement and death transitions and an instrumental-variable version continue to support the positive effect.
- The baseline sample contains 29,384 observations and 3,598 firms.

Economic message: Table I establishes the core result: the same firm innovates more under a more trusting CEO.

TABLE I

BASELINE EFFECT OF A CEO'S TRUST ON THEIR FIRM'S PATENTS

Dependent variable	arsinh (future patent applications)							
Specification	Baseline spec. (1)	Add. ind. trends (2)	Forward cites (3)	Patent value (4)	Retired or died (5)	Retired or died IV (6)	Country controls (7)	Other traits (8)
CEO's trust	0.062***	0.066***	0.099***	0.115***	0.085*		0.042**	0.054**
	(0.017)	(0.017)	(0.029)	(0.041)	(0.043)		(0.017)	(0.023)
Post × change in trust						0.082*		
						(0.046)		
Firm and year fixed effects	X	X	X	X	X	X	X	X
Baseline controls	X	X	X	X	X	X	X	X
Industry × year fixed effects			X					
Home country controls							X	
Other cultural traits								X
Observations	29,384	29,384	29,384	20,218	3,758	3,758	29,384	29,384
Firms	3,598	3,598	3,598	3,168	374	374	3,598	3,598

Notes. This table reports the baseline effect of a CEO's inherited trust on their firm's patents using equation (2). The baseline sample includes all observations of firm f × year t × its current CEO d. The dependent variable is the inverse hyperbolic sine of firm f's patent-application count in year $t+1$. The explanatory variable is CEO d's inherited-trust measure, standardized by its standard deviation at the ethnicity level. Baseline controls include (i) the firm's age, age squared, and (ii) the CEO's age, age squared, gender, education dummies, and tenure in the firm. Columns (3) and (4) dependent variables weight patent applications by their forward citations (column (3)) and patent values (i.e., firm f's excess stock return on patent grant date, estimated by Kogan et al. 2017) (column (4)). Columns (5) and (6) focus on CEO transitions in which the predecessors retire between the age of 63 and 67 or die in or within a year of leaving the office. Column (6) further uses the predecessor's trust measure as an instrument for the subsequent change in the CEO's trust (i.e., the difference between the successor's and predecessor's standardized trust measures); both instrument and instrumented variables are interacted with a post-transition period dummy. The corresponding first-stage coefficient (standard error) and Kleibergen-Paap Wald F-statistic are −0.940 (0.030) and 957.43, respectively. Column (7) additionally controls for the CEO's home country's macroeconomic characteristics, including ln(GDP), ln(population), GDP growth rate, high school graduate share, average percentile ranking of World Bank governance indices, ln(U.S. exports + U.S. imports), and ln(total patent applications filed at the country's patent office). Column (8) additionally controls for the CEO's other inherited cultural traits, including work ethic, risk preference, time preference (i.e., patience), and the CEO's ethnic group's high-income share. (See Online Appendix C for further details.) Standard errors in parentheses are clustered by the CEO's main ethnicity. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE II

THE CEO'S TRUST EFFECT BY INDUSTRY AND THE CEO'S BACKGROUND

Dependent variable	arsinh(future patent applications)						
Sample	Pharma/chemical ind.		ICT/electronic ind.		Other ind.	Full sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CEO's trust	0.076**	0.012	0.095**	0.137**	0.036**		
	(0.035)	(0.049)	(0.046)	(0.053)	(0.016)		
Trust × tenure as CEO		0.025*		−0.014			
		(0.014)		(0.012)			
Trust × MBA/no grad. deg.						0.094***	
						(0.022)	
Trust × non-MBA grad. deg.						−0.009	
						(0.036)	
Trust × no prior R&D exp.							0.064***
							(0.015)
Trust × prior R&D exp.							0.001
							(0.133)
Firm and year fixed effects	X	X	X	X	X	X	X
Baseline controls	X	X	X	X	X	X	X
Observations	5,234	5,234	7,274	7,274	16,876	29,384	29,384
Firms	675	675	862	862	2,061	3,598	3,598

Notes. This table explores the heterogeneous effects of a CEO's trust on their firm's patents by industry and the CEO's background using equation (2). The baseline sample includes all observations of firm f × year t × its current CEO d. The dependent variable is the inverse hyperbolic sine of firm f's patent-application count in year $t+1$. The explanatory variable is CEO d's GSS-based inherited-trust measure, standardized by its standard deviation at the ethnicity level. Baseline controls include (i) the firm's age, age squared, and (ii) the CEO's age, age squared, gender, education dummies, and tenure in the firm. Columns (1) and (2) use the subsample of firms in pharmaceutical and chemical industries (BoardEx sectors "Chemicals," "Pharmaceuticals and Biotechnology," and "Health"). Columns (3) and (4) use the subsample of firms in ICT and electronic industries (BoardEx sectors "Electronic & Electrical Equipment," "Information Technology Hardware," and "Software & Computer Services"). Column (5) uses the subsample of firms in the remaining industries. Columns (6) and (7) interact the CEO's trust measure with a dummy indicating whether the CEO has a non-MBA master's or a doctorate degree (column (6)), or whether the CEO has prior R&D experience (column (7)). Standard errors in parentheses are clustered by the CEO's main ethnicity. * $p < .10$, ** $p < .05$, *** $p < .01$.

5.4 Table II: heterogeneity by industry and CEO background

Read:

- The CEO trust effect is positive in pharmaceutical and chemical industries and in ICT/electronic industries.
- It is smaller in other industries, consistent with trust being more important where innovation is central and risky.
- The effect is stronger for CEOs without prior R&D experience than for CEOs with prior R&D experience.
- The effect is also related to CEO education categories, suggesting that the relevance of trust depends on what information or expertise the CEO already has.

Economic message: Trust matters most when innovative activity is central and when the CEO must rely on trust rather than direct technical expertise.

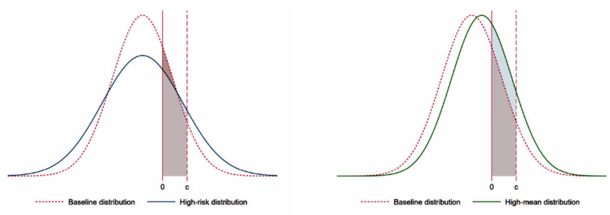
5.5 Figures IV and V plus Table III: quality distribution and exploration

Read:

- Figure IV illustrates two possible mechanisms: a rightward shift of project quality and a mean-preserving spread that creates more high-quality outcomes.
- Figure V shows that the effect of CEO trust is concentrated in the top patent-quality deciles.

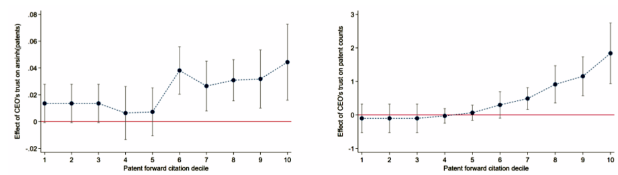
- Table III confirms the quality channel: CEO trust increases explorative patents, disruptive patents, patent importance, scientific citations, technological scope, and average patent value.
- The coefficient is 0.064 for explorative patents, 0.044 for disruptive patents, and 0.099 for patent importance.
- CEO trust also raises the share of explorative patents and average forward citations.

Economic message: The evidence supports the exploration mechanism. Trust does not merely increase patent quantity; it changes the distribution toward more valuable and more exploratory innovation.



A. Risk-taking mechanism   B. Mean-shifting mechanism

Figure IV: project-quality distributions



A. IHS patent counts   B. Patent counts

Figure V: effect by patent-quality decile

TABLE III

CEOs' Trust's Effect on Quality-Weighted Patents

Dependent variable	arsinh(future quality-weighted patents)					Share	arsinh	arsinh
Quality measure	Explorative pat. (1)	Disruptive pat. (2)	Patent importance (3)	Backward NPL cites (4)	Tech. scope (5)	(explor. pat.) (6)	(avg. cites) (7)	(avg. value) (8)
CEO's trust	0.064***	0.044***	0.099***	0.102***	0.060**	0.011**	0.043**	0.058**
	(0.013)	(0.015)	(0.026)	(0.030)	(0.023)	(0.004)	(0.019)	(0.024)
Firm and year fixed effects	X	X	X	X	X	X	X	X
Baseline controls	X	X	X	X	X	X	X	X
Observations	29,384	29,384	29,384	29,384	29,284	29,384	29,384	20,218
Firms	3,598	3,598	3,598	3,598	3,598	3,598	3,598	3,168

Notes. This table reports the effect of CEOs' trust on quality-weighted patents using equation (2). The baseline sample includes all observations of firm f × year t × its current CEO d. The dependent variable is the inverse hyperbolic sine of firm f's patent-application count in year $t+1$, weighted by explorative patents (i.e., patents mostly citing knowledge that is new, as developed by Benner and Tushman 2002) (column (1)); disruptive patents (i.e., patents that destabilize the status quo, as developed by Funk and Owen-Smith 2017) (column (2)); patent importance (i.e., the ratio between patent-text-based measures of patent impact and patent novelty, computed by Kelly et al. 2021) (column (3)); backward citations to non-patent (i.e., scientific) literature (column (4)); and patent technological scope (column (5)) (details are in Online Appendix A.2). The dependent variable in column (6) is the share of explorative patents; column (7) the inverse hyperbolic sine of the average forward citations per patent; and column (8) the inverse hyperbolic sine of the average patent value (i.e., firm f's excess stock return on patent grant date, estimated by Kogan et al. 2017); all computed among patents filed by firm f in year $t+1$ (or set to zero if firm f has no patent application in year $t+1$). The explanatory variable is CEO d's GSS-based inherited-trust measure, standardized by its standard deviation at the ethnicity level. Baseline controls include (i) the firm's age, age squared, and (ii) the CEO's age, age squared, gender, education dummies, and tenure in the firm. Standard errors in parentheses are clustered by the CEO's main ethnicity. * $p < .10$, ** $p < .05$, *** $p < .01$.

5.6 Table IV: CEO trust and corporate trust culture

Read:

- Table IV uses employee review text classified by a fine-tuned BERT model.
- CEO trust is positively associated with overall trust sentiment in employee reviews.
- The effect is especially strong for top-down trust: managers' trust toward workers.
- The bottom-up trust effect, workers' trust toward managers, is much smaller and statistically weaker.
- For R&D workers, the top-down trust effect is larger, consistent with the innovation mechanism.
- Dynamic specifications show that trust culture strengthens over the CEO term.

Economic message: More trusting CEOs appear to change the internal culture of the firm, especially by increasing managers' willingness to trust workers.

5.7 Table V: CEO trust and director selection

Read:

- More trusting CEOs appoint more trusting new directors.
- New directors who stay through the CEO term are especially likely to be more trusting.

TABLE IV

CEO'S TRUST AND CORPORATE TRUST CULTURE

Dependent variable	Employee review's trust sentiment						
Trust direction	Both directions		Top down	Bottom up	Top down		
Sample	All reviews		All reviews		R&D workers	All reviews	R&D workers
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CEO's trust	0.251**	0.371	0.343**	0.052	0.776***		
	(0.095)	(0.277)	(0.160)	(0.080)	(0.259)		
Trust × $\mathbb{1}_{Tenure\ as\ CEO \leqslant 2}$						0.316**	0.691**
						(0.136)	(0.268)
Trust × $\mathbb{1}_{Tenure\ as\ CEO \in [3,4]}$						0.345**	0.869***
						(0.142)	(0.248)
Trust × $\mathbb{1}_{Tenure\ as\ CEO \in [5,6]}$						0.396**	1.146***
						(0.146)	(0.227)
Trust × $\mathbb{1}_{Tenure\ as\ CEO \in [7,8]}$						0.413**	1.307***
						(0.191)	(0.342)
Trust × $\mathbb{1}_{Tenure\ as\ CEO \geqslant 9}$						0.277	1.286**
						(0.228)	(0.582)
Positive reviews	0.194	0.194	0.133	0.029	0.216	0.133	0.216
Negative reviews	1.253	1.253	0.513	0.342	0.636	0.513	0.636
Normalized effect size	0.173	0.257	0.531	0.141	0.911		

TABLE IV

CONTINUED

Dependent variable	Employee review's trust sentiment						
Trust direction	Both directions		Top down	Bottom up	Top down		
Sample	All reviews		All reviews		R&D workers	All reviews	R&D workers
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Baseline controls	X	X	X	X	X	X	X
Industry and review year FEs	X						
Firm and review year FEs		X	X	X	X	X	X
Observations	331,091	331,091	331,091	331,091	24,763	331,091	24,763
Firms	266	266	266	266	122	266	122

Notes. This table presents the effect of a CEO's trust on online employee reviews' trust sentiment. Each observation is a review, and the sample includes all CEO terms (firm f × CEO d) with at least 50 reviews during 2008–2017. Columns (5) and (7) further restrict the sample to CEO terms with at least 25 reviews by R&D workers. The dependent variable takes the value 100, −100, or 0 if the review reflects positive trust, negative trust, or no information on trust. Columns (1) and (2) consider both directions of trust; columns (3), (5), (6), and (7) consider only top-down trust (i.e., managers' trust toward workers); and column (4) considers only bottom-up trust (i.e., workers' trust toward managers). The explanatory variable is CEO d's inherited-trust measure, standardized by its standard deviation at the ethnicity level. Baseline controls include (i) the firm's age, age squared; (ii) the CEO's age, age squared, gender, education dummies, non-U.S. dummy, and tenure in the firm; (iii) the CEO's other cultural traits, including work ethic, risk preference, time preference (i.e., patience), and the CEO's ethnic group's high-income share (details are in Online Appendix C); and (iv) review characteristics, including CEO approval, current job dummy, and time gap between review year and last year of employment. Columns (6) and (7) interact the CEO's trust with time-period dummies. Positive and negative reviews report the percentage shares of relevant positive and negative trust reviews in the corresponding sample. Normalized effect size is the ratio between the coefficient on CEO's trust and the total percentage share of positive and negative trust reviews. Each observation is weighted by the inverse of the corresponding CEO term's review count in the respective estimation sample. Standard errors in parentheses are clustered by the CEO's main ethnicity. * $p < .10$, ** $p < .05$, *** $p < .01$.

- Directors who leave under more trusting CEOs tend to be less trusting.
- The average trust of the full board rises under more trusting CEOs.

Economic message: Trust can cascade through the organization through personnel selection. CEOs may build trust culture by selecting directors and managers whose beliefs are aligned with their own.

TABLE V

CEO'S TRUST AND DIRECTOR SELECTION

Dependent variable	Individual director's trust			Average directors' trust			
Director sample	New	Staying	Leaving	New	Staying	Leaving	All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CEO's trust	0.029*	0.037**	−0.047*	0.041**	0.053***	−0.044	0.009**
	(0.016)	(0.016)	(0.025)	(0.017)	(0.019)	(0.032)	(0.004)
Observation unit	Director × firm × year × CEO			Firm × year × CEO			
Firm and year FEs	X	X	X	X	X	X	X
Baseline controls	X	X	X	X	X	X	X
Observations	17,218	14,428	10,752	10,430	9,293	7,756	29,266
Firms	3,363	3,347	2,862	3,363	3,347	2,862	3,555

Notes. This table reports the association between CEOs' trust and directors' trust. Columns (1)–(3): samples include observations of director r × firm f × year t × its current CEO d over the baseline 2000–2011 period, separately for newly appointed directors (i.e., new directors, column (1)), newly appointed directors who stay on the board until the end of the CEO's term (i.e., staying directors, column (2)), and directors who leave the board the following year and before the end of the CEO's term (i.e., leaving directors, column (3)). The dependent variable is director r's inherited-trust measure. Columns (4)–(7): samples include observations of firm f × year t × its current CEO d. The dependent variable is the average inherited-trust measure among all new directors (column (4)), all staying directors (column (5)), all leaving directors (column (6)), and all directors (column (7)). Directors who ever serve as the firm's CEO are always excluded. The explanatory variable is CEO d's inherited-trust measure. Baseline controls include (i) the firm's age, age squared, and (i) the CEO's age, age squared, gender, education dummies, and tenure in the firm. Standard errors in parentheses are clustered by the CEO's main ethnicity. * $p < .10$, ** $p < .05$, *** $p < .01$.

5.8 Table VI: the right amount of trust

Read:

- The trust effect is stronger when the pre-transition researcher pool is higher quality.
- The interaction between trust and researcher quality is positive and statistically significant for all patents, top-quality patents, and R&D efficiency.
- In quintile specifications, the effect is strongest in the top researcher-quality quintiles.
- Trust also has heterogeneous real effects: sales, employment, and TFP effects depend on the quality of the researcher pool.

Economic message: Trust is productive only when directed toward a trustworthy environment. High trust can be valuable in a good research organization but costly if it prevents screening out low-quality researchers.

TABLE VI

CEOs' Trust's Effect by Pre-transition Researcher-Pool Quality

Forward	One-year				Two-year		
Dependent variable: future	All patents (1)	Top qual. patents (2)	R&D efficiency (3)	All patents (4)	ln(sales) (5)	ln(employment) (6)	TFP (7)
CEO's trust	0.063^{***}	0.055^{***}	0.038^{***}		-0.036^{**}	-0.035^{***}	0.004
	(0.017)	(0.015)	(0.013)		(0.017)	(0.013)	(0.013)
Trust × researcher quality	0.045^{**}	0.044^{**}	0.028^{**}		0.045^{***}	0.035^{***}	0.013
(in pretransition period)	(0.018)	(0.019)	(0.011)		(0.014)	(0.010)	(0.011)
Trust × quality quintile 1				0.015			
				(0.039)			
Trust × quality quintile 2				0.032			
				(0.027)			
Trust × quality quintile 3				0.064			
				(0.050)			
Trust × quality quintile 4				0.084^{***}			
				(0.029)			
Trust × quality quintile 5				0.130^{***}			
				(0.039)			

TABLE VI

Continued

Forward	One-year				Two-year		
Dependent variable: future	All patents (1)	Top qual. patents (2)	R&D efficiency (3)	All patents (4)	ln(sales) (5)	ln(employment) (6)	TFP (7)
Event and year fixed effects	X	X	X	X	X	X	X
Baseline controls	X	X	X	X	X	X	X
Observations	19,506	19,506	19,506	19,506	17,873	17,684	17,125
Events	2,278	2,278	2,278	2,278	2,229	2,216	2,168

Notes. This table explores the heterogeneous effects of CEOs' trust on firms' patents by pre-transition researcher-pool quality using equation (2) and the sample constructed from CEO transition events (details are in Online Appendix A.4). For each event, I include all firm f × year t × its current CEO d observations that correspond to the predecessor's and successor's terms. The explanatory variable is CEO d's inherited-trust measure, standardized by its standard deviation at the ethnicity level. The firm-level proxy for researcher-pool quality is computed from averaging the residuals from regressing patents on observable firm and CEO characteristics, controlling for two-digit SIC industry and year fixed effects over the $(-1, 0)$ two-year pre-transition window. The dependent variables in columns (1)–(4) are the inverse hyperbolic sine of firm f's patent-application count (columns (1) and (4)), high-quality (i.e., above-median) patent-application count (column (2)), and R&D efficiency, calculated as patent-application count over lagged R&D expenditure (column (3)), in year $t+1$. The dependent variables in columns (5)–(7) are firm f's ln(sales) (column (5)), ln(employment) (column (6)), and TFP computed from value added, employment, and capital following Olley and Pakes (1996) (column (7)). Baseline controls include (i) the firm's age, age squared, arsinh(R&D expenditure), and (ii) the CEO's age, age squared, gender, education dummies, and tenure in the firm. Column (4) interacts the CEO's trust measure with researcher-pool quality quintile dummies (computed based on firm-level proxy for pre-transition researcher-pool quality). The remaining columns interact the CEO's trust measure with the firm-level proxy for pre-transition pool quality. Standard errors in parentheses are clustered by the CEO's main ethnicity. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

5.9 Table VII: within-CEO bilateral-trust effect

Read:

- Table VII uses CEO fixed effects and compares how the same CEO's bilateral trust differs across inventor countries.
- A one-standard-deviation increase in CEO bilateral trust raises corresponding inventor-country patenting by about 5% in the baseline specification.
- The effect survives additional fixed effects, including firm-by-inventor-country fixed effects.
- Citation-weighted patenting produces a larger coefficient, around 9%.
- Results are robust to excluding same-country CEO–inventor pairs and controlling for language, geography, and genetic distance.

Economic message: The bilateral-trust design absorbs all fixed CEO traits. It shows that even within the same CEO, inventors in countries toward which the CEO has more trust produce more patents.

TABLE VII

The CEO's Trust Effect in the Bilateral-Trust Sample

Dependent variable	arsinh(future patent applications)							
Specification	Baseline	Add. fixed effects		Fwd. cites	Country pairwise controls			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CEO's bilateral trust	0.052^{**}	0.051^{**}	0.034^{*}	0.095^{**}	0.047^{*}	0.042^{*}	0.052^{**}	0.052^{**}
	(0.023)	(0.024)	(0.021)	(0.039)	(0.029)	(0.025)	(0.024)	(0.024)
Common language dummy						0.057		
						(0.047)		
Geographical distance (1,000 km)							−0.002	
							(0.019)	
Genetic distance (z-score)								0.018
								(0.084)

TABLE VII

Continued

Dependent variable	arsinh(future patent applications)							
Specification	Baseline	Add. fixed effects		Fwd. cites	Country pairwise controls			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CEO fixed effects	X		X	X	X	X	X	X
Inventor country fixed effects	X			X	X	X	X	X
Firm × year fixed effects	X	X		X	X	X	X	X
CEO × year fixed effects		X						
Inv. country × year fixed effects		X						
Firm × inv. country fixed effects			X					
Year fixed effects			X					
Excl. same-country pairs					X			
Observations	23,284	23,284	23,284	23,284	20,878	23,284	23,284	22,881
Firm × inv. country	3,481	3,481	3,481	3,481	3,145	3,481	3,481	3,421
Firms	730	730	730	730	496	730	730	728

Notes. This table reports the effect of the CEO's bilateral trust toward a country on patents by inventors from that country using equation (4). The sample includes all observations of firm f × year t × its current CEO d × country c such that firm f has patents by inventors from country c during 2000–2012. An inventor's country is inferred from their patent-listed address for non-U.S.-based inventors. The dependent variable is firm f's total patent-application count by inventors from country c in year $t+1$. The explanatory variable is CEO d's bilateral trust toward individuals from country c, standardized by its standard deviation at the country-pair level. Column (4)'s dependent variable weights patent applications by their forward citations. Column (5) excludes same-country CEO–inventor country pairs. Columns (6)–(8) control for CEO–inventor country pairwise linguistic, geographical, and genetic distances, including whether the countries share a common language (column (6)), the weighted geographical distance between the countries (column (7)), and the weighted genetic distance between the countries' populations (Spolaore and Wacziarg 2016) (column (8)). Standard errors in parentheses are clustered by the CEO's main ethnicity × inventor country. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

6 Short summary

- The paper provides a firm-level micro-foundation for the link between trust and growth.
- It constructs a matched CEO–firm–patent dataset covering 5,753 CEOs, 3,598 firms, and about 700,000 patents.
- CEO inherited trust is inferred from last names, historical censuses, and ethnic-origin trust in the GSS.
- The main design uses CEO transitions with firm fixed effects.
- A one-standard-deviation increase in CEO trust is associated with about 6% more future patents.
- The effect is concentrated in high-quality, explorative, disruptive, and valuable patents.
- The mechanism is tolerance of failure and higher researcher risk-taking.

- Employee-review text shows that trusting CEOs improve trust culture, especially top-down trust.
- Director-selection evidence shows that more trusting CEOs select more trusting directors.
- Trust is most effective when researcher-pool quality is high.
- A within-CEO bilateral-trust design shows that inventors in countries trusted more by the CEO produce more patents.

7 Conclusion

7.1 Core conclusion

Nguyen shows that CEO trust is an economically important driver of corporate innovation. Firms patent more after switching to a more trusting CEO, and this increase is concentrated in higher-quality innovation. The evidence supports the view that trust affects growth through firm-level culture and exploratory R&D incentives.

7.2 Economic intuition

Innovation requires risk-taking under uncertainty. Researchers often fail even when they are talented and exert effort. A trusting CEO is more likely to interpret failure as bad luck rather than bad type, which makes managers more tolerant of failed experimentation. Anticipating this tolerance, researchers are more willing to pursue high-risk, high-upside projects.

7.3 Takeaway

The paper turns trust from a macro-level cultural variable into a micro-level managerial trait with measurable firm consequences. Trust helps innovation when it supports exploratory research and when the research organization is sufficiently high quality. It is therefore both a source of innovative capability and a managerial resource that must be matched to the right environment.